

Ultrasonic Module HC-SR04 Interfacing with 8051



Crystal
Oscillator

HC - SR04

Vcc

Trig

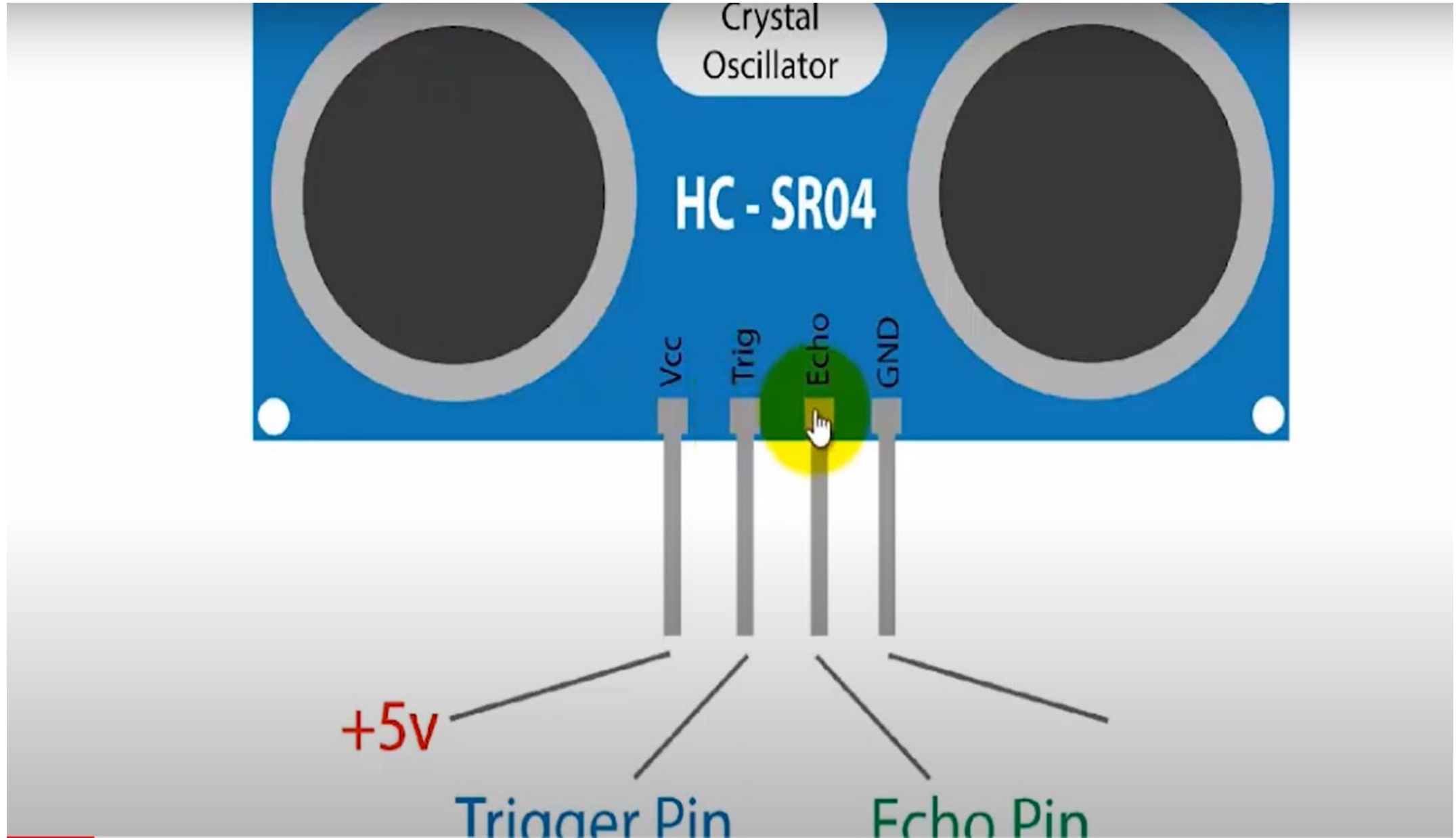
Echo

GND

+5v

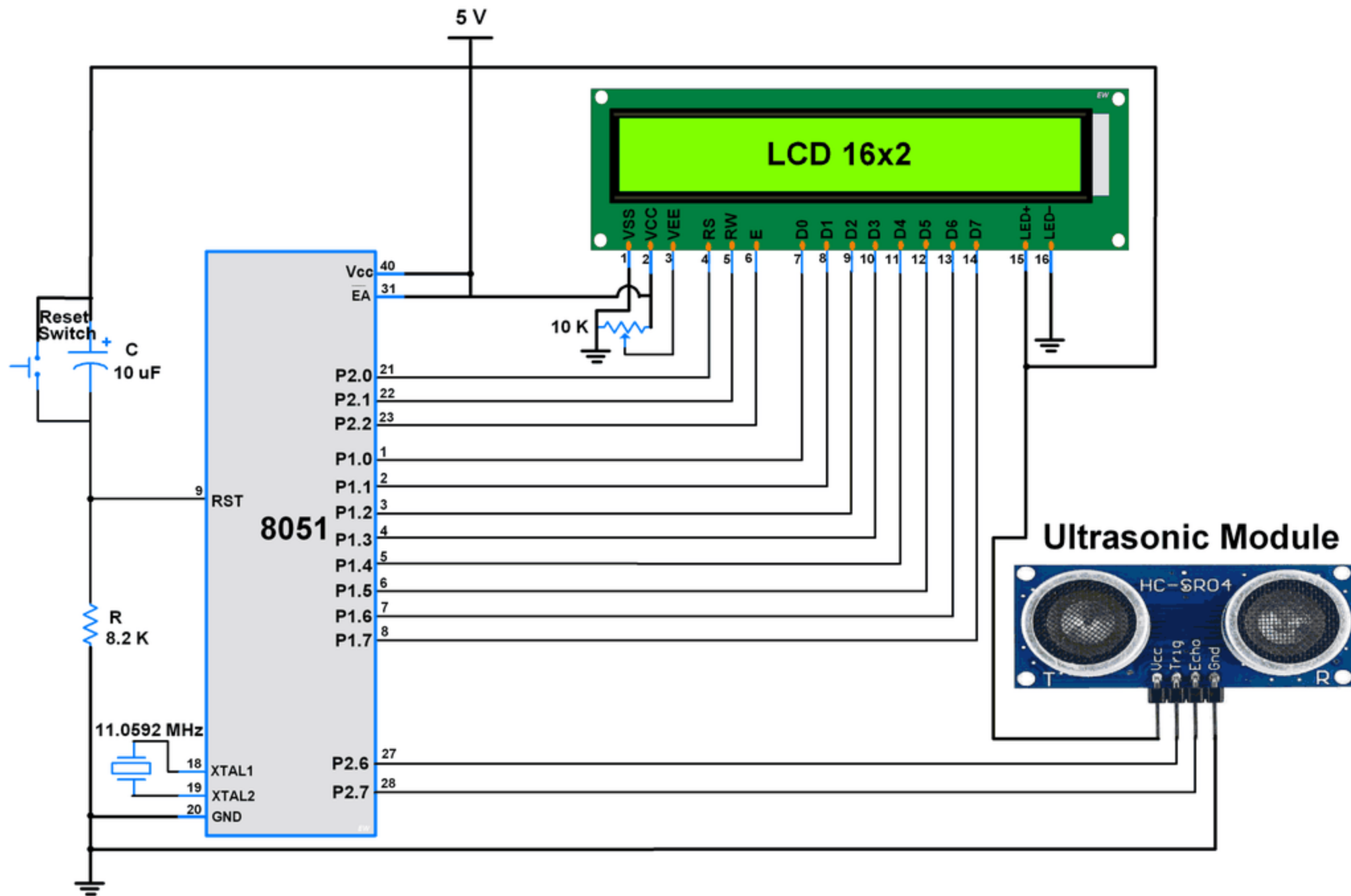
Trigger Pin

Echo Pin



Ultrasonic Module HC-SR04 Interfacing with 8051

- Ultrasonic Module HC-SR04 works on the principle of SONAR and RADAR system.
- The HC-SR04 module has an ultrasonic transmitter, receiver, and control circuit on a single board.
- The module has only 4 pins, Vcc, Gnd, Trig, and Echo.
- When a pulse of $10\mu\text{sec}$ or more is given to the Trig pin, 8 pulses of 40 kHz are generated. After this, the Echo pin is made high by the control circuit in the module.
- The echo pin remains high till it gets the echo signal of the transmitted pulses back.
- The time for which the echo pin remains high, i.e. the width of the Echo pin gives the time taken for generated ultrasonic sound to travel towards the object and return.
- Using this time and the speed of sound in air, we can find the distance of the object using a simple formula for distance using speed and time.



Steps of Programming

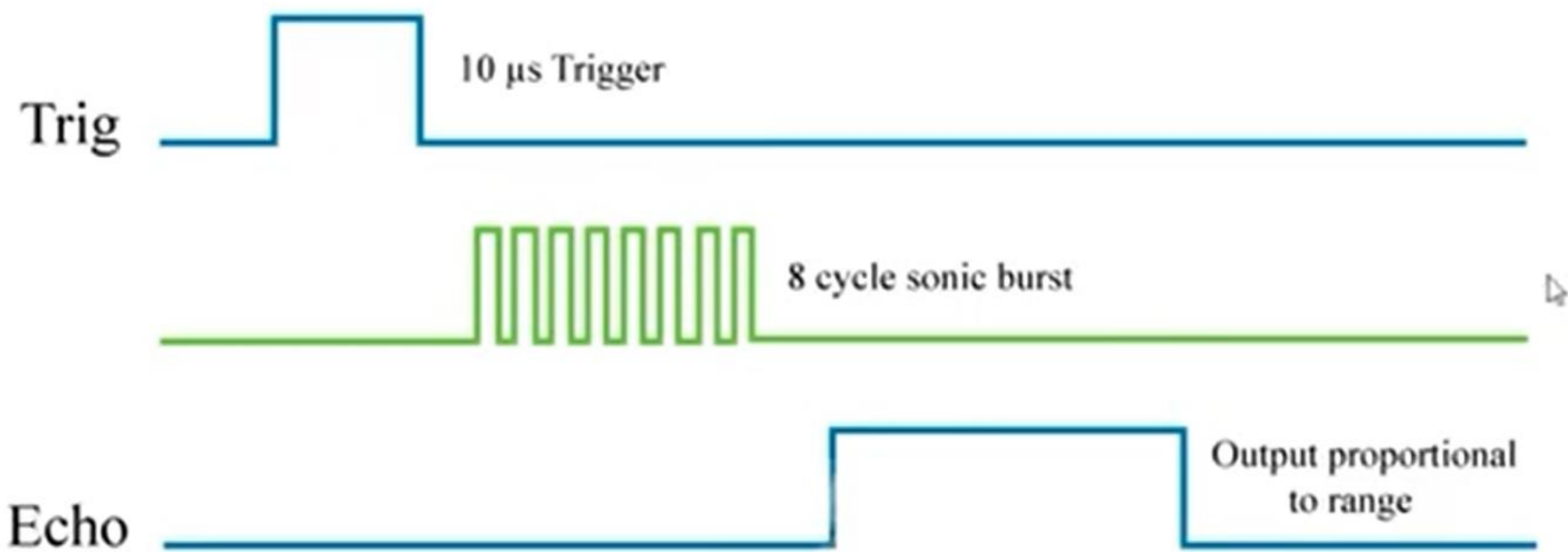
1. 8051 microcontroller needs to transmit at least 10 μ s trigger pulse to the HC-SR04 Trig Pin.
2. After getting a trigger pulse, HC-SR04 automatically sends eight 40 kHz sound waves and waits for rising edge output at the Echo pin.
3. When the rising edge capture occurs at the Echo pin which is connected to the input of 8051, start Timer of 8051 and again wait for the falling edge on the Echo pin.
4. As soon as the falling edge is captured at the Echo pin, the microcontroller reads the count of the Timer. This time count is used to calculate the distance to an object.

Calculation (distance in cm)

- Where,
-

$$Distance = \frac{SoundVelocity * Time}{2}$$

- Sound Velocity = 34300 (in cm per second)
- Here, the oscillator frequency of AT89S52 (8051) is 11.0592 MHz then the timer frequency of 8051 will be 921.6 kHz. So, the time required to execute 1 instruction is 1.085 us.
- So, the timer gets incremented after 1.085 us time elapse.



PROGRAM

```
MOV DPTR,#LUT      // moves the address of LUT to DPTR
MOV P1,#00000000B  // sets P1 as output port
MOV P0,#00000000B  // sets P0 as output port
CLR P3.0           // sets P3.0 as output for sending trigger
SETB P3.1          // sets P3.1 as input for receiving echo
MOV TMOD,#00100000B // sets timer1 as mode 2 auto reload timer
MAIN: MOV TL1,#207D // loads the initial value to start counting from
      MOV TH1,#207D // loads the reload value
      MOV A,#00000000B // clears accumulator
      SETB P3.0      // starts the trigger pulse
      ACALL DELAY1   // gives 10uS width for the trigger pulse
      CLR P3.0       // ends the trigger pulse
HERE: JNB P3.1,HERE  // loops here until echo is received
BACK: SETB TR1       // starts the timer1
HERE1: JNB TF1,HERE1 // loops here until timer overflows (ie;48 count)
      CLR TR1        // stops the timer
      CLR TF1        // clears timer flag 1
      INC A          // increments A for every timer1 overflow
      JB P3.1,BACK    // jumps to BACK if echo is still available
      MOV R4,A        // saves the value of A to R4
      ACALL DLOOP     // calls the display loop
      SJMP MAIN       // jumps to MAIN loop
```

PROGRAM

```
DELAY1: MOV R6,#2D    // 10uS delay
LABEL1: DJNZ R6,LABEL1
        RET
DLOOP:  MOV R5,#100D   // loads R5 with 100D
BACK1:  MOV A,R4       // loads the value in R4 to A
        MOV B,#100D    // loads B with 100D
        DIV AB         // isolates the first digit
        SETB P1.0      // activates LED display unit D1
        ACALL DISPLAY  // calls DISPLAY subroutine
        MOV P0,A       // moves digit drive pattern for 1st digit to P0
        ACALL DELAY    // 1mS delay
        ACALL DELAY
        MOV A,B        // moves the remainder of 1st division to A
        MOV B,#10D     // loads B with 10D
        DIV AB         // isolates the second digit
        CLR P1.0       // deactivates LED display unit D1
        SETB P1.1      // activates LED display unit D2
        ACALL DISPLAY
        MOV P0,A       // moves digit drive pattern for 2nd digit to P0
        ACALL DELAY
```

PROGRAM

ACALL DELAY

MOV A,B // moves the remainder of 2nd division to A

CLR P1.1 // deactivates LED display unit D2

SETB P1.2 // activates LED display unit D3

ACALL DISPLAY

MOV P0,A // moves the digit drive pattern for 3rd digit to P0

ACALL DELAY

ACALL DELAY

CLR P1.2 // deactivates LED display unit D3

DJNZ R5,BACK1 // repeats the display loop 100 times

RET

DELAY: MOV R7,#250D // 1mS delay

LABEL2: DJNZ R7,LABEL2

RET

DISPLAY: MOVC A,@A+DPTR // gets the digit drive pattern for the content in A

CPL A // complements the digit drive pattern (see Note 1)

RET

LUT: DB 3FH // look up table (LUT) starts here

DB 06H

DB 5BH